Same-Zero Gaussian–Phase Coupling for Riemann-Zeta Zeros

Víctor Casaña-Giner[1*]

[1]Universidad Nacional de Educación a Distancia, C/ Bravo Murillo, 38, 28015, Madrid, Spain
ORCID: 0000-0002-8252-2608.

Corresponding author(s). E-mail(s): vcasana1@alumno.uned.es;

Abstract

Let γ_k denote the positive ordinates of the nontrivial zeros of the Riemann zeta function, counted with multiplicity. Write $M(T) = \frac{T}{2\pi}\log\frac{T}{2\pi e} + \frac{7}{8}$, define the smooth Riemann–von Mangoldt clock by $M(\tau_k) = k - \frac{1}{2}$, and set $\sigma_k = \sqrt{2\log\log\tau_k}/\log(\tau_k/(2\pi))$ and $X_k = (\gamma_k - \tau_k)/\sigma_k$. Building on the Gaussian fluctuation theorem for X_k, we prove a same-zero metric coupling: for every $J \geq 1$, for Lebesgue-almost every fixed $u \in \mathbb{R}^J$,

$$(X_k, (u_1\gamma_k, \ldots, u_J\gamma_k) \bmod 2\pi) \Longrightarrow (G, \Theta),$$

where G is standard Gaussian, Θ is Haar-uniform on $\mathbb{T}^J$, and the two factors are independent. The proof uses an exact sinc-kernel L^2 identity, local zero density, uniform moment bounds, Borel–Cantelli, and moment–Fourier determination. A first-order expansion then gives the corresponding Gaussian–phase product law. We write R_{SF} for the radial law of four independent uniform phases. We show that a deterministic zeta-zero observable converges to the "shark-fin" law R_{SF}. It is noteworthy that this classical radial law emerges through Hlawka–Weyl equidistribution in a deterministic zeta-zero setting, where no random-walk structure of the zeros is assumed or known. In the classical literature, the same law is obtained from genuinely stochastic constructions based on independent uniform steps and random walks. We also obtain an adjacent-zero diagonal consequence and identify the finer increment scale on which nearest-neighbor spacing survives.

Keywords: Riemann zeta function, zeta zeros, Gaussian fluctuations, equidistribution, metric independence, zero spacings

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1 Introduction

Most classical zero statistics are naturally formulated in a height parameter T. This is not accidental: zero counting, density estimates, pair correlation, and fixed-frequency phase averages are usually expressed over ordinates $0 < \gamma \le T$. For a statistic attached to the individual zero, however, the ordered sequence viewpoint is complementary. Once the k-th zero is compared with its smooth Riemann–von Mangoldt clock position, the same index k also carries deterministic phase coordinates $u_j\gamma_k$. The statistic studied here is precisely this same-index object.

The height and ordinal viewpoints are asymptotically linked by Riemann–von Mangoldt inversion, but they emphasize different statistics.

There are three mathematically distinct objects in this paper.

First, Kargin's normalized displacement X_k is a one-zero statistic comparing the ordinate γ_k with its smooth Riemann–von Mangoldt clock position τ_k.

Second, the vector

$$(u_1\gamma_k, \dots, u_J\gamma_k) \bmod 2\pi$$

is also a one-zero statistic. Hlawka's theorem [3] gives fixed-frequency equidistribution of individual zeta-zero phases; refinements and simultaneous fractional-part results include [4–6]. The new point developed below is the asymptotic independence, for almost every frequency vector, between these two structures at the same index.

Third, nearest-neighbor spacing is intrinsically a two-zero statistic; the classical pair-correlation and random-matrix context is represented by [1, 2]. It depends on (γ_k, γ_{k+1}), equivalently on (X_k, X_{k+1}) at a scale below the first-order marginal limit. The fixed-frequency four-phase shark-fin law does not contain this adjacent-zero information.

The first main theorem concerns the interaction between the first two one-zero statistics; the adjacent-zero material is a separate consequence of the same ordinal viewpoint.

The first two objects are both attached to a single ordered zero, but they come from different traditions. Kargin's theorem describes the fluctuation of the k-th ordinate around its smooth clock position. Hlawka-type phase theorems describe the distribution of fixed-frequency functions of the ordinates. The question addressed here is whether these two pieces of information remain asymptotically coupled when evaluated at the same index.

The existing marginal theorems do not decide this question: phase equidistribution alone does not control the displacement weight X_k^ℓ, and Kargin's Gaussian marginal alone contains no fixed-frequency torus information. In particular, the issue is not torus equidistribution alone, but twisted equidistribution with the displacement weights X_k^ℓ.

The proof mechanism is an exact sinc-kernel L^2 identity, followed by local zero-density control, Borel–Cantelli, and moment–Fourier determination. The shark-fin law is historically motivating but logically secondary; the adjacent-zero results are included as a second consequence of the ordered-zero viewpoint.

The law R_{SF} is classical as a four-phase radial law, but its appearance here is a deterministic zeta-zero realization. The four phases are fixed functions of a single ordinate γ_k; their Haar product structure is obtained only in the limiting empirical sense through Hlawka's theorem and Weyl's criterion. Thus the theorem is neither a

random-walk model for the zeros nor a restatement of the short-walk construction. The Gaussian–phase product law arising from the same-zero coupling is a different distribution.

2 Notation and conventions

Unless otherwise stated, empirical convergence is with respect to the uniform prefix measure on the relevant zero indices.

Symbol	Meaning
$\rho_k = \beta_k + i\gamma_k$	k-th nontrivial zero in the upper half-plane, ordered by ordinate and counted with multiplicity.
$N(T)$	Zero-counting function for ordinates $0 < \gamma \le T$.
$M(T)$	Smooth Riemann–von Mangoldt counting term used to define the clock.
$S(T)$	Argument term in the Riemann–von Mangoldt formula.
τ_k	Smooth clock position associated with the k-th zero by (3.1).
ε_k	Ordinate displacement $\gamma_k - \tau_k$.
$\sigma_k, \widetilde{\sigma}_k$	Normalizing scales for the displacement; $X_k = \varepsilon_k/\sigma_k$.
X_k	Normalized displacement of the k-th zero from its smooth clock position.
G	Standard Gaussian random variable.
$\mathbb{T}^J$, Θ	J-torus and a Haar-uniform torus variable.
$u = (u_1, \dots, u_J)$	Fixed real frequency vector for zero phases.
$\eta = (\eta_1, \dots, \eta_J)$	Fixed coefficient vector in exponential residuals.
$m \in \mathbb{Z}^J$	Integer Weyl/Fourier frequency.
$I_{N,\ell}(v)$	Twisted moment average $N^{-1}\sum X_k^\ell e^{iv\gamma_k}$.
$\mathcal{V}, \mathcal{U}_J$	Full-measure sets of admissible scalar and vector frequencies.
$E_k(u,\eta), W_k(u,\eta)$	Zero–clock exponential residual and its first-order phase factor.
$\Delta_k(u)$	Deterministic four-phase two-chord observable whose radial limit is the shark-fin law.
$R_{\mathrm{SF}}, p_{\mathrm{SF}}$	Shark-fin probability law and its density; Theorem 7.4 realizes it deterministically along zeta zeros.
K_N	Uniformly chosen index in the indicated prefix/window.
r_N	Length of a growing consecutive-zero window.
L_k	$\log(\tau_k/(2\pi))$.
d_k^{clk}	Local clock spacing $2\pi/L_k$.
D_k	Microscopic displacement increment $\sqrt{\log\log\tau_k}(X_{k+1} - X_k)$.
G_k^{gap}	Nearest-neighbor gap normalized by $2\pi/L_k$.
ν, m_ℓ	Generic limiting real law and its moments in the decoupling principle.
$\mathbb{E}, \mathbb{P}, \mathrm{Var}, \mathrm{Corr}$	Expectation, probability, variance, and correlation.
$\operatorname{sinc} x$	$\sin x/x$, with $\operatorname{sinc} 0 = 1$.

3 Zeros, clock, and Kargin's theorem

Definition 3.1 (Zeros and smooth clock) Let $\rho_k = \beta_k + i\gamma_k$ denote the nontrivial zeros with $\gamma_k > 0$, counted with multiplicity and ordered by increasing ordinate. Define the smooth Riemann–von Mangoldt term

$$M(T) := \frac{T}{2\pi} \log \frac{T}{2\pi e} + \frac{7}{8}.$$

Fix k_0 so that for every $k \geq k_0$ the quantity τ_k is the unique solution of

$$M(\tau_k) = \frac{\tau_k}{2\pi} \log \frac{\tau_k}{2\pi e} + \frac{7}{8} = k - \frac{1}{2}. \tag{3.1}$$

Define

$$\varepsilon_k := \gamma_k - \tau_k, \qquad \sigma_k := \frac{\sqrt{2 \log\log \tau_k}}{\log(\tau_k/(2\pi))}, \qquad X_k := \frac{\varepsilon_k}{\sigma_k}. \tag{3.2}$$

For $k < k_0$, define $X_k = 0$.

In all expressions involving X_k^0, the finite initial segment is interpreted separately. Equivalently, all averages may be taken over $k_0 \leq k \leq N$; this changes prefix empirical measures by $O(k_0/N)$.

Kargin's displayed normalization uses

$$\widetilde{\sigma}_k = \frac{\sqrt{2 \log\log \tau_k}}{\log \tau_k}.$$

Since

$$\frac{\widetilde{\sigma}_k}{\sigma_k} = \frac{\log(\tau_k/(2\pi))}{\log \tau_k} \longrightarrow 1,$$

the two normalizations are asymptotically equivalent. When passing from a pointwise scale ratio to prefix averages, the finite head is separated explicitly.

Theorem 3.2 (Kargin [7]) *If $1/2 < \theta \leq 1$ and k is uniform in*

$$[N, N + \lfloor N^\theta \rfloor - 1] \cap \mathbb{Z},$$

then

$$\frac{\gamma_k - \tau_k}{\widetilde{\sigma}_k} \Longrightarrow G,$$

where $G \sim N(0,1)$. Every fixed integer moment converges to the corresponding Gaussian moment.

Lemma 3.3 (Pointwise growth) *For all sufficiently large k,*

$$|X_k| \ll \log(2 + k).$$

Proof The Riemann–von Mangoldt formula in the classical Backlund form gives $N(T) = M(T) + S(T) + O(1/T)$ with $S(T) \ll \log(2 + T)$ [9, Ch. IX]. Here

$$M'(T) = \frac{1}{2\pi} \log \frac{T}{2\pi}.$$

Since $M(\tau_k) = k - \frac{1}{2}$, the mean-value theorem converts an $O(\log T)$ error in zero count into an $O(1)$ displacement in height, because $M'(T) \asymp \log T$. In particular,

$$|\gamma_k - \tau_k| \ll \frac{\log \tau_k}{M'(\tau_k)} \ll 1,$$

while

$$\sigma_k \asymp \frac{\sqrt{\log\log \tau_k}}{\log \tau_k}.$$

Hence $|X_k| \ll \log \tau_k / \sqrt{\log\log \tau_k} \ll \log(2+k)$. □

Lemma 3.4 (Prefix moments) *For every integer $r \geq 0$,*

$$\frac{1}{N} \sum_{k \leq N} X_k^r \longrightarrow \mu_r := \mathbb{E} G^r.$$

For every real $r \geq 0$ there exists $C_r < \infty$ such that

$$\frac{1}{N} \sum_{k \leq N} |X_k|^r \leq C_r$$

for all N.

Proof Apply Theorem 3.2 with $\theta = 1$ on dyadic blocks $[M, 2M)$. A prefix is decomposed into disjoint dyadic blocks plus a bounded initial remainder. Large blocks have moments arbitrarily close to μ_r, while the fixed head contributes $O(M_0/N)$. Any rounding or endpoint correction contains only $O(1)$ indices and is $o(1)$ after division by N by Lemma 3.3. Higher even moments and Hölder give the absolute-moment bounds. After increasing C_r to absorb the finite range $N < N_r$, the bound holds for all N.

For the change from $\widetilde{\sigma}_k$ to σ_k, fix M_0. On $k \geq M_0$ the scale ratio is uniformly close to one once M_0 is large, while $k \leq M_0$ contributes $O(M_0/N)$. This transfers the prefix moment limits to (3.2). □

4 Metric cancellation of twisted moments

For $\ell \in \mathbb{Z}_{\geq 0}$ and $v \in \mathbb{R}$, define

$$I_{N,\ell}(v) := \frac{1}{N} \sum_{k_0 \leq k \leq N} X_k^\ell e^{iv\gamma_k}.$$

The normalization by N, rather than by $N - k_0 + 1$, is used throughout; the two empirical normalizations are asymptotically equivalent.

Lemma 4.1 (Local density and sinc row bound) *Let $T_N = \gamma_N$. Uniformly in $k \leq N$ and $r \geq 0$,*

$$\#\{k' \leq N : |\gamma_{k'} - \gamma_k| \leq r\} \ll (1+r)\log(2+T_N).$$

Hence for each fixed $A > 0$,

$$\sup_{k \leq N} \sum_{k' \leq N} |\mathrm{sinc}(A(\gamma_k - \gamma_{k'}))| \ll_A \log^2(2+T_N).$$

Proof The Riemann–von Mangoldt formula with the classical $O(\log T)$ error gives

$$N(T+1) - N(T) \ll \log(2+T),$$

as in Titchmarsh, Theorem 9.2 [9]. Cover the interval centered at γ_k by unit intervals. For the sinc sum use

$$|\operatorname{sinc}(Ax)| \le \min\left(1, \frac{1}{A|x|}\right)$$

and sum over unit shells. The harmonic shell sum contributes the second logarithm. □

Theorem 4.2 (Metric twisted-moment cancellation) *There exists a full-Lebesgue-measure set $\mathcal{V} \subset \mathbb{R}$, with $0 \notin \mathcal{V}$, such that for every $v \in \mathcal{V}$ and every integer $\ell \ge 0$,*

$$I_{N,\ell}(v) \longrightarrow 0.$$

Proof Fix ℓ and $A > 0$. Direct integration gives the exact identity

$$\frac{1}{2A}\int_{-A}^{A} |I_{N,\ell}(v)|^2\, dv = \frac{1}{N^2} \sum_{k_0 \le k,k' \le N} X_k^{\ell} X_{k'}^{\ell} \operatorname{sinc}(A(\gamma_k - \gamma_{k'}))\,. \tag{4.1}$$

Using

$$|X_k^{\ell} X_{k'}^{\ell}| \le \frac{1}{2}\left(|X_k|^{2\ell} + |X_{k'}|^{2\ell}\right),$$

Lemma 4.1, symmetry, Lemma 3.4, and $\log T_N \ll \log N$,

$$\frac{1}{2A}\int_{-A}^{A} |I_{N,\ell}(v)|^2\, dv \ll_{\ell,A} \frac{\log^2 N}{N}.$$

Take $N_j = j^2$. Chebyshev and Borel–Cantelli give $I_{N_j,\ell}(v) \to 0$ for almost every v.

If $M = j^2 \le N < (j+1)^2$ and $d = N - M = O(\sqrt{M})$, then

$$|I_{N,\ell}(v) - I_{M,\ell}(v)| \le \left|\frac{1}{N} - \frac{1}{M}\right| \sum_{k_0 \le k \le M} |X_k|^{\ell} + \frac{1}{N} \sum_{M < k \le N} |X_k|^{\ell}.$$

The first term is $O_\ell(N^{-1/2})$. Cauchy–Schwarz and the 2ℓ-moment bound give the second term as $O_\ell(N^{-1/4})$, uniformly in v. Intersect over integer $A > 0$ and $\ell \ge 0$. □

5 Moment–Fourier determination and same-zero coupling

Lemma 5.1 (Moment–Fourier determination) *Let (Y_N, Θ_N) be probability measures on $\mathbb{R} \times \mathbb{T}^J$. Assume uniformly bounded even moments of every fixed order. Let ν be a moment-determinate probability law on $\mathbb{R}$, with all moments finite, and assume explicitly that the algebraic polynomials are dense in $L^2(\nu)$. Write $m_\ell = \int x^\ell\, d\nu(x)$. If*

$$\mathbb{E}_N\left(Y_N^{\ell} e^{im\cdot\Theta_N}\right) \longrightarrow m_\ell \mathbf{1}_{m=0} \tag{5.1}$$

for every $\ell \ge 0$ and $m \in \mathbb{Z}^J$, then

$$(Y_N, \Theta_N) \Longrightarrow (Y, \Theta), \tag{5.2}$$

where $Y \sim \nu$, Θ is Haar-uniform on $\mathbb{T}^J$, and the two factors are independent.

Proof Even-moment bounds give tightness in the real coordinate, while $\mathbb{T}^J$ is compact. Let λ be a subsequential limit. Uniform integrability passes every mixed polynomial–Fourier moment to λ: for a monomial of degree ℓ, a uniformly bounded sufficiently high even moment (for example the 2ℓ-th moment when $\ell \geq 1$) gives the required uniform integrability. The $m = 0$ moments and moment determinacy identify the real marginal as ν.

Disintegrate $\lambda(dx, d\theta) = \nu(dx)\lambda_x(d\theta)$. For every nonzero $m \in \mathbb{Z}^J$ and every polynomial P,

$$\int_{\mathbb{R}} P(x) \left(\int_{\mathbb{T}^J} e^{im\cdot\theta} \lambda_x(d\theta) \right) \nu(dx) = 0. \tag{5.3}$$

The explicit polynomial-density hypothesis in $L^2(\nu)$ forces the inner Fourier coefficient to vanish for ν-almost every x. Intersecting over the countable set of nonzero m, λ_x has all nontrivial torus Fourier coefficients equal to zero for ν-almost every x; hence λ_x is Haar measure. Every subsequential limit is therefore $\nu \otimes m_{\mathbb{T}^J}$. □

Remark 5.2 (Polynomial density) For the principal application ν is Gaussian, and polynomial density in $L^2(\nu)$ follows from the Hermite polynomial basis; see, for example, [21, Chs. V, IX]. In the general decoupling principle below, polynomial density is retained as an explicit hypothesis; no implication from moment determinacy alone to unweighted $L^2(\nu)$-density is used. For classical moment-problem background, see [19].

Define

$$\mathcal{U}_J := \bigcap_{m \in \mathbb{Z}^J \setminus \{0\}} \{u \in \mathbb{R}^J : m \cdot u \in \mathcal{V}\}.$$

By Fubini, $\mathcal{U}_J$ has full Lebesgue measure. In particular, every $u \in \mathcal{U}_J$ has rationally independent coordinates: if $m \in \mathbb{Z}^J \setminus \{0\}$ satisfied $m \cdot u = 0$, then the definition of $\mathcal{U}_J$ would force $0 \in \mathcal{V}$, contrary to Theorem 4.2.

Theorem 5.3 (Same-zero Gaussian–phase coupling) *For every $J \geq 1$ and every $u \in \mathcal{U}_J$, under the uniform prefix measure,*

$$(X_k, (u_1\gamma_k, \ldots, u_J\gamma_k) \bmod 2\pi) \Longrightarrow (G, \Theta), \tag{5.4}$$

where G is standard Gaussian, Θ is Haar-uniform on $\mathbb{T}^J$, and the two components are independent. More precisely, for every $\ell \geq 0$ and $m \in \mathbb{Z}^J$,

$$\frac{1}{N} \sum_{k_0 \leq k \leq N} X_k^\ell e^{i(m\cdot u)\gamma_k} \longrightarrow \mu_\ell \mathbf{1}_{m=0}.$$

Thus every mixed polynomial–Fourier moment converges.

Proof For $m \in \mathbb{Z}^J$ and $\ell \geq 0$,

$$\frac{1}{N} \sum_{k_0 \leq k \leq N} X_k^\ell e^{i(m\cdot u)\gamma_k} \longrightarrow \mu_\ell \mathbf{1}_{m=0}.$$

For $m = 0$ this is Lemma 3.4; for $m \neq 0$ it is Theorem 4.2. The uniformly bounded even moments of every fixed order required by Lemma 5.1 are supplied by the second assertion of Lemma 3.4. Apply Lemma 5.1. □

Remark 5.4 (Novel content) Theorem 5.3 is not Hlawka's phase equidistribution and is not Kargin's Gaussian marginal. Its content is the same-index asymptotic product coupling between the normalized zero displacement and the torus phase vector. This is the principal new theorem of the paper.

5.1 A reusable metric decoupling lemma

Proposition 5.5 (Metric phase-decoupling principle) *Let $(\Psi_k)_{k\ge1}$ be a real sequence such that, for every integer $r \ge 0$,*

$$\frac{1}{N}\sum_{k\le N}\Psi_k^r \longrightarrow m_r, \tag{5.5}$$

and for every real $r \ge 0$,

$$\sup_{N\ge1}\frac{1}{N}\sum_{k\le N}|\Psi_k|^r < \infty. \tag{5.6}$$

Assume that (m_r) determines a probability law ν on $\mathbb{R}$, that ν has all moments finite, and that the algebraic polynomials are dense in $L^2(\nu)$. Then for every $J \ge 1$ there is a full-Lebesgue-measure set $\mathcal{U}_J(\Psi) \subset \mathbb{R}^J$ such that, for every $u \in \mathcal{U}_J(\Psi)$,

$$(\Psi_k, (u_1\gamma_k, \dots, u_J\gamma_k) \bmod 2\pi) \Longrightarrow (Y, \Theta), \tag{5.7}$$

where $Y \sim \nu$, Θ is Haar-uniform on $\mathbb{T}^J$, and Y and Θ are independent.

Proof For fixed $\ell \ge 0$ repeat the sinc-kernel argument of Theorem 4.2 with X_k^ℓ replaced by Ψ_k^ℓ. The only input from the weight is the uniform prefix bound for the 2ℓ-th absolute moment in (5.6); hence

$$\frac{1}{2A}\int_{-A}^{A}\left|\frac{1}{N}\sum_{k\le N}\Psi_k^\ell e^{iv\gamma_k}\right|^2 dv \ll_{\ell,A} \frac{\log^2 N}{N}. \tag{5.8}$$

Borel–Cantelli on $N_j = j^2$ and the same interpolation argument give, for almost every $v \ne 0$,

$$\frac{1}{N}\sum_{k\le N}\Psi_k^\ell e^{iv\gamma_k} \longrightarrow 0. \tag{5.9}$$

Intersect over ℓ and over the countable Weyl frequencies $m \cdot u$, $m \in \mathbb{Z}^J \setminus \{0\}$. The untwisted moments converge by (5.5); moment determinacy, the explicit polynomial-density hypothesis, and Lemma 5.1 yield the product law (5.7). □

Corollary 5.6 (Clock phases) *For the same u,*

$$(X_k, (u_1\tau_k, \dots, u_J\tau_k) \bmod 2\pi) \Longrightarrow (G, \Theta).$$

Proof Since $\varepsilon_k = \sigma_k X_k$, $\sigma_k \to 0$, and X_k is tight, $\varepsilon_k \to 0$ in probability. Slutsky applies to the torus coordinates. □

6 Gaussian–phase product law

For fixed $u, \eta \in \mathbb{R}^J$ define

$$E_k(u,\eta) := \sum_{j=1}^{J} \eta_j \left(e^{iu_j\gamma_k} - e^{iu_j\tau_k} \right), \qquad W_k(u,\eta) := i \sum_{j=1}^{J} \eta_j u_j e^{iu_j\tau_k}.$$

Lemma 6.1 (Exact first-order control) *For every k,*

$$|E_k - \varepsilon_k W_k| \le \frac{\varepsilon_k^2}{2} \sum_{j=1}^{J} |\eta_j| u_j^2.$$

Consequently, for every fixed $q \ge 1$,

$$\frac{E_k}{\sigma_k} - X_k W_k \longrightarrow 0$$

in L^q under the uniform prefix measure.

Proof Use $|e^{iy} - 1 - iy| \le y^2/2$. After division by σ_k, the remainder is bounded by a constant times $\sigma_k X_k^2$. Lemma 3.4 supplies the required uniform moments. □

Theorem 6.2 (Metric Gaussian–phase product law) *For every $J \ge 1$, $u \in \mathcal{U}_J$, and fixed $\eta \in \mathbb{R}^J$,*

$$\frac{|E_k(u,\eta)|}{\sigma_k} \Longrightarrow |G| \left| \sum_{j=1}^{J} \eta_j u_j e^{i\Theta_j} \right|.$$

Every fixed moment converges.

Proof By Corollary 5.6,

$$(X_k, (u_1\tau_k, \dots, u_J\tau_k) \bmod 2\pi) \Longrightarrow (G, \Theta).$$

Since

$$W_k(u,\eta) = i \sum_{j=1}^{J} \eta_j u_j e^{iu_j\tau_k}$$

is a bounded continuous function of the clock-phase vector, the continuous mapping theorem gives

$$X_k W_k(u,\eta) \Longrightarrow iG \sum_{j=1}^{J} \eta_j u_j e^{i\Theta_j}.$$

Lemma 6.1 gives

$$\frac{E_k(u,\eta)}{\sigma_k} - X_k W_k(u,\eta) \longrightarrow 0$$

in every fixed L^q. Slutsky's theorem and continuity of the modulus yield the stated weak limit. For moment convergence, W_k is uniformly bounded and Lemma 3.4 supplies moments of X_k of arbitrarily high order; hence the relevant powers are uniformly integrable. The same L^q-remainder control transfers those moments from $|X_k W_k|$ to $|E_k|/\sigma_k$. □

Corollary 6.3 (One frequency) *For $J = 1$, every $u_1 \neq 0$, and every real η_1,*

$$\frac{|E_k|}{\sigma_k} \Longrightarrow |\eta_1 u_1|\,|G|.$$

Proof For $J = 1$, $|W_k| = |\eta_1 u_1|$ identically. Use Lemma 3.4, Gaussian moment determinacy, and the first-order control. □

6.1 Finite-height product diagnostic

As a finite-height diagnostic for Theorem 6.2, use all 2,001,052 entries of the canonical `zeross.bin` and `zLamb.bin` arrays, with

$$u = (1, \sqrt{2}), \qquad \eta = (1, 1),$$

one obtains:

Quantity	Value
count	2,001,052
$\langle X \rangle$	-3.48×10^{-7}
$\langle X^2 \rangle$	0.848887862
skewness	-1.03×10^{-5}
kurtosis	2.765506
Gaussian kurtosis	3
$\langle \lvert W\rvert^2 \rangle$	3.000001
exact $\mathbb{E}\lvert W\rvert^2$	3
$\langle X^2 \lvert W\rvert^2 \rangle$	2.546416
$\langle \lvert E\rvert^2/\sigma^2 \rangle$	2.512254
$\mathrm{Corr}(X^2, \lvert W\rvert^2)$	-1.10×10^{-4}
$\mathrm{Corr}(\lvert X\rvert, \lvert W\rvert)$	3.66×10^{-5}

The finite-height X distribution is visibly platykurtic as well as having variance below one; rescaling its variance alone does not make its fourth and sixth moments Gaussian. Figure 1 displays the corresponding full-data radial comparison.

7 The deterministic four-phase shark-fin law

Definition 7.1 (Shark-fin law) Let $m_{\mathbb{T}^4}$ denote Haar probability measure on $\mathbb{T}^4$, and let $\Theta = (\Theta_1, \ldots, \Theta_4) \sim m_{\mathbb{T}^4}$. We denote by R_{SF} the probability law of

$$R_{\mathrm{SF}}(\Theta) := \left| e^{i\Theta_1} + e^{i\Theta_2} + e^{i\Theta_3} + e^{i\Theta_4} \right|.$$

Its radial density is

$$p_{\mathrm{SF}}(r) = r \int_0^\infty J_0(rx) J_0(x)^4 \, x \, dx, \qquad 0 < r < 4,$$

and its first two moments are

$$\mathbb{E}R_{\mathrm{SF}} = 1.799092479842851\ldots, \qquad \mathbb{E}R_{\mathrm{SF}}^2 = 4.$$

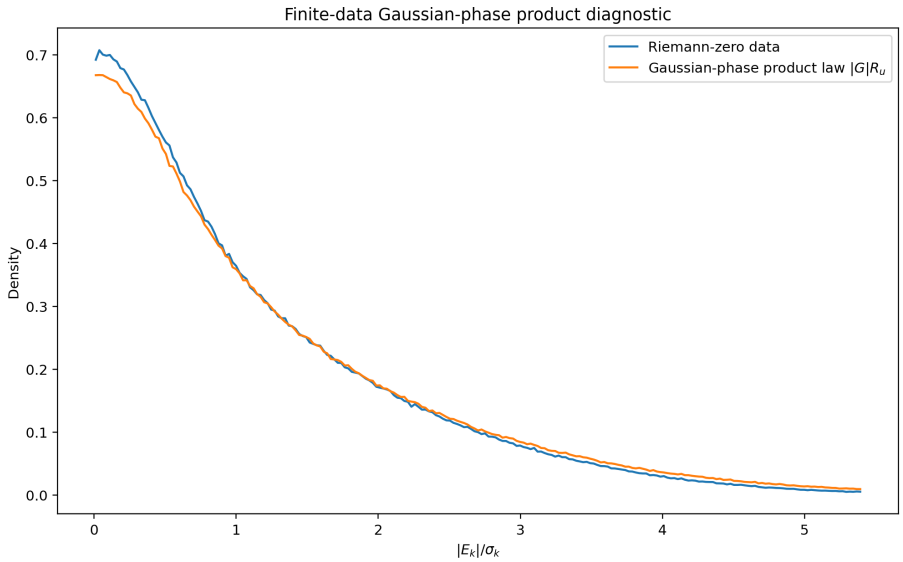


Fig. 1 Finite-height distribution of the normalized zero–clock exponential residual compared with the Gaussian–phase product reference law.

Lemma 7.2 (Two-chord representation) *Let*

$$A = e^{i\alpha_2} - e^{i\alpha_1}, \qquad B = e^{i\beta_2} - e^{i\beta_1},$$

with four Haar-uniform phase coordinates. Then

$$|A - B| \stackrel{d}{=} R_{\mathrm{SF}}.$$

Proof

$$A - B = e^{i\alpha_2} + e^{i(\alpha_1+\pi)} + e^{i(\beta_2+\pi)} + e^{i\beta_1}.$$

Adding π preserves Haar measure. □

Theorem 7.3 (Deterministic four-phase criterion) *If*

$$(f_1(k), f_2(k), f_3(k), f_4(k)) \bmod 2\pi$$

is equidistributed in $\mathbb{T}^4$ *and*

$$\Delta_k = (e^{if_1(k)} - e^{if_2(k)}) - (e^{if_3(k)} - e^{if_4(k)}),$$

then

$$|\Delta_k| \Longrightarrow R_{\mathrm{SF}}.$$

Proof Equidistribution means that the empirical measures of $(f_1(k), \dots, f_4(k)) \bmod 2\pi$ converge weakly to Haar probability measure on $\mathbb{T}^4$. The map

$$(\theta_1, \theta_2, \theta_3, \theta_4) \longmapsto \left|(e^{i\theta_1} - e^{i\theta_2}) - (e^{i\theta_3} - e^{i\theta_4})\right|$$

is continuous. The continuous mapping theorem therefore gives convergence of the empirical radial laws to the Haar law of this two-chord expression, which is R_{SF} by Lemma 7.2. □

Theorem 7.4 (Deterministic four-phase shark-fin law along zeta zeros) *Let* $u = (u_1, u_2, u_3, u_4) \in \mathbb{R}^4$ *have rationally independent coordinates and define*

$$\Delta_k(u) = (e^{iu_1\gamma_k} - e^{iu_2\gamma_k}) - (e^{iu_3\gamma_k} - e^{iu_4\gamma_k}).$$

Then

$$|\Delta_k(u)| \Longrightarrow R_{\mathrm{SF}}.$$

Proof By multidimensional Weyl's criterion [13], the phase vector is equidistributed in $\mathbb{T}^4$ iff

$$\frac{1}{N}\sum_{k\le N} e^{i(m\cdot u)\gamma_k} \to 0$$

for every $m \in \mathbb{Z}^4 \setminus \{0\}$. Rational independence gives $m \cdot u \neq 0$. Hlawka's theorem [3] supplies the one-dimensional Weyl limit for every fixed nonzero real frequency. Apply Theorem 7.3. □

Remark 7.5 (Reference law and zeta-zero realization) The law R_{SF} is classical as a four-phase radial law. Its occurrence in Theorem 7.4, however, is a deterministic zeta-zero realization. The four phases are fixed functions of γ_k, and their Haar product structure is obtained from Hlawka's theorem and Weyl's criterion. In the short-walk literature the same density is derived from four independent uniform steps [10–12]; that random-walk construction is not an assumption of the zeta-zero theorem. The noteworthy point here is that the same radial law arises without assuming or knowing any random-walk structure of the zeros. The distinct index dynamics are quantified in Proposition 7.6 and in Remark 7.7.

Proposition 7.6 (Phase-window rigidity) *Fix* $J \ge 1$ *and* $u = (u_1, \dots, u_J) \in \mathbb{R}^J$, *and put*

$$\Theta_k(u) := (u_1\gamma_k, \dots, u_J\gamma_k) \bmod 2\pi.$$

Let r_N *be any nonnegative integer sequence with* $r_N = o(\log N)$, *and let* K_N *be uniform on* $\{k_0, \dots, N - r_N\}$. *For any fixed metric* $d_{\mathbb{T}^J}$ *induced by the Euclidean metric on* $\mathbb{R}^J$,

$$\max_{0\le j\le r_N} d_{\mathbb{T}^J}\big(\Theta_{K_N+j}(u), \Theta_{K_N}(u)\big) \longrightarrow 0 \tag{7.1}$$

in L^1, *hence in probability.*

In particular, for a rationally independent four-frequency vector u, *let* $R_k(u) := |\Delta_k(u)|$. *If* $j_N \le r_N$, *then*

$$(R_{K_N}(u), R_{K_N+j_N}(u)) \Longrightarrow (R_{\mathrm{SF}}, R_{\mathrm{SF}}),$$

where the two coordinates on the right are the same random variable, and

$$\mathrm{Corr}(R_{K_N}(u), R_{K_N+j_N}(u)) \longrightarrow 1. \tag{7.2}$$

Thus the one-point limiting law is R_{SF}, *whereas the index process is locally rigid on every* $o(\log N)$ *window.*

Proof By zero ordering,

$$\max_{0\le j\le r_N} |\gamma_{K_N+j} - \gamma_{K_N}| = \gamma_{K_N+r_N} - \gamma_{K_N}.$$

Writing $M_N = N - r_N - k_0 + 1$, telescoping gives

$$\begin{aligned}\mathbb{E}(\gamma_{K_N+r_N} - \gamma_{K_N}) &= \frac{1}{M_N}\sum_{k=k_0}^{N-r_N}(\gamma_{k+r_N} - \gamma_k)\\ &= \frac{\sum_{m=N-r_N+1}^{N}\gamma_m - \sum_{m=k_0}^{k_0+r_N-1}\gamma_m}{M_N}\\ &\le \frac{r_N\gamma_N}{M_N}.\end{aligned}$$

Riemann–von Mangoldt inversion gives $\gamma_N \asymp N/\log N$, hence the last quantity is $O(r_N/\log N) = o(1)$. Since quotient distance on the torus is no larger than Euclidean distance of representatives,

$$\max_{0\le j\le r_N} d_{\mathbb{T}^J}\big(\Theta_{K_N+j}(u), \Theta_{K_N}(u)\big) \le \|u\|_2\big(\gamma_{K_N+r_N} - \gamma_{K_N}\big),$$

which proves (7.1) in L^1.

For the four-phase observable, the map

$$\theta \longmapsto \left|(e^{i\theta_1} - e^{i\theta_2}) - (e^{i\theta_3} - e^{i\theta_4})\right|$$

is Lipschitz on $\mathbb{T}^4$. Therefore $R_{K_N+j_N}(u) - R_{K_N}(u) \to 0$ in probability, uniformly for $j_N \le r_N$. The variables are bounded by 4, so the convergence also holds in every fixed L^p. Replacing the full prefix $\{k \le N\}$ by $\{k \le N - r_N\}$ does not change the limiting empirical law, since $r_N = o(N)$. Therefore Theorem 7.4 gives $R_{K_N}(u) \Rightarrow R_{\mathrm{SF}}$, whose variance is positive; the L^2 convergence of the difference then gives (7.2). □

Remark 7.7 (Measured index correlations and the finite-height regime) Proposition 7.6 is asymptotic in the height, and its limiting correlation-one regime is not numerically visible at the heights of the present dataset. It is useful to record the finite-height behaviour, both because it gives a direct diagnostic of the indexed dynamics and because it must not be read as contradicting the proposition.

For $u = (1, \sqrt{2}, \sqrt{3}, \sqrt{5})$ and $R_k(u) = |\Delta_k(u)|$, the empirical index correlations over the full dataset are

lag	1	2	3	4
$\mathrm{Corr}(R_k, R_{k+\mathrm{lag}})$	0.8911	0.6520	0.3557	0.0796
lag	5	7	10	20
$\mathrm{Corr}(R_k, R_{k+\mathrm{lag}})$	−0.1245	−0.2800	−0.2619	−0.1329

The full table of twenty lags is supplied in `outputs/shark_fin_index_correlations.txt`.

The exact phase increment is

$$\Theta_{k+1}(u) - \Theta_k(u) = u\,(\gamma_{k+1} - \gamma_k) \pmod{2\pi}.$$

For scale comparison only, replace the fluctuating zero gap by the local clock spacing $d_k^{\mathrm{clk}} = 2\pi/L_k$. The corresponding clock-scale phase increment is

$$\delta\Theta_k^{\mathrm{clk}}(u) := \frac{2\pi}{L_k}(u_1, \dots, u_4). \tag{7.3}$$

At the top of the dataset,

$$\|\delta\Theta_k^{\mathrm{clk}}(u)\|_2 \approx \sqrt{11} \times 0.5192 \approx 1.72 \quad \text{radians.}$$

Thus the small-displacement regime used asymptotically in Proposition 7.6 is not yet a small one-step scale here. The fastest coordinate has a clock-scale 2π-revolution length $L/u_{\max} \approx 12.10/2.236 \approx 5.41$ indices, comparable with the observed first sign change between lags 4 and 5. This comparison is a finite-height diagnostic, not a pointwise approximation asserting that individual zeta-zero gaps equal their clock spacings.

The negative lobe is equally informative: the correlation reaches about -0.280 near lag 7 and then returns toward zero with oscillation. Independent repeated realizations of the classical four-step construction would have zero correlation between distinct samples; the classical one-point law itself does not prescribe an index process. The measured profile instead shows deterministic oscillatory index structure, consistent with the quasi-periodic geometry of the four fixed phases. This concerns the indexed process, not the one-point law: by Theorem 7.4, the limiting empirical law of $R_k(u)$ is R_{SF}, while Proposition 7.6 proves local rigidity in the asymptotic $o(\log N)$ regime. Equality of the one-point law with a classical radial distribution is therefore not an independence statement across zero indices.

7.1 Fixed-frequency numerical audit

Using all 2,001,052 ordinates and

$$u = (1, \sqrt{2}, \sqrt{3}, \sqrt{5}),$$

the fixed-frequency four-phase observable gives:

Quantity	Value
sample count	2,001,052
mean radius, data	1.799094395840
exact shark-fin mean	1.799092479843
mean deviation	1.915997×10^{-6}
second moment, data	4.000000861075
exact second moment	4
second-moment deviation	8.610750×10^{-7}
approx. KS shape diagnostic vs 8×10^6 reference draws	3.822×10^{-4}

The two moment deviations are the cleaner diagnostics. For orientation only, an i.i.d. four-step sample of this size would have standard error

$$\sqrt{\frac{4 - (\mathbb{E}R_{\mathrm{SF}})^2}{N}} = 6.18 \times 10^{-4}$$

for the sample mean; this is not used as a sampling model for the deterministic zeta-zero sequence.

The Landau–Gonek comparison used here is unconditional and does not rely on a finite verification of RH. Away from prime-power logarithmic resonances, the Landau–Gonek explicit formula supplies the natural comparison scale for the corresponding finite-height Weyl sums [16, 17]. Odlyzko identifies `zeros6.txt` as the first 2,001,052

zeros [22]. Platt–Trudgian [20] has a different and independent role: their rigorous verification of RH through height $3 \cdot 10^{12}$ lies far above the largest ordinate in the present table, about 1.13×10^6, so every nontrivial zero in the height range used here lies on the critical line. Thus the critical-line status of the entire numerical range is independently certified; it is not an assumption behind the Landau–Gonek comparison. These finite checks are provenance checks only and are not used in the proof of Theorem 7.4.

The second moment has an exact decomposition. With $\epsilon = (+1, -1, -1, +1)$,

$$|\Delta_k(u)|^2 = 4 + 2 \sum_{1 \le a < b \le 4} \epsilon_a \epsilon_b \cos\big((u_a - u_b)\gamma_k\big),$$

so

$$\frac{1}{N} \sum_{k \le N} |\Delta_k(u)|^2 - 4$$

is exactly a linear combination of six fixed-frequency Weyl sums.

Nonresonance check. For $u = (1, \sqrt{2}, \sqrt{3}, \sqrt{5})$, the four coordinates are rationally independent and every difference $u_a - u_b$ is a nonzero algebraic number. If such a difference were $m \log p$, then $e^{u_a - u_b} = p^m$ would be algebraic, contradicting Lindemann–Weierstrass; see [18]. Thus none of the six difference frequencies is a prime-power logarithmic resonance. The full-data comparison is shown in Figure 2.

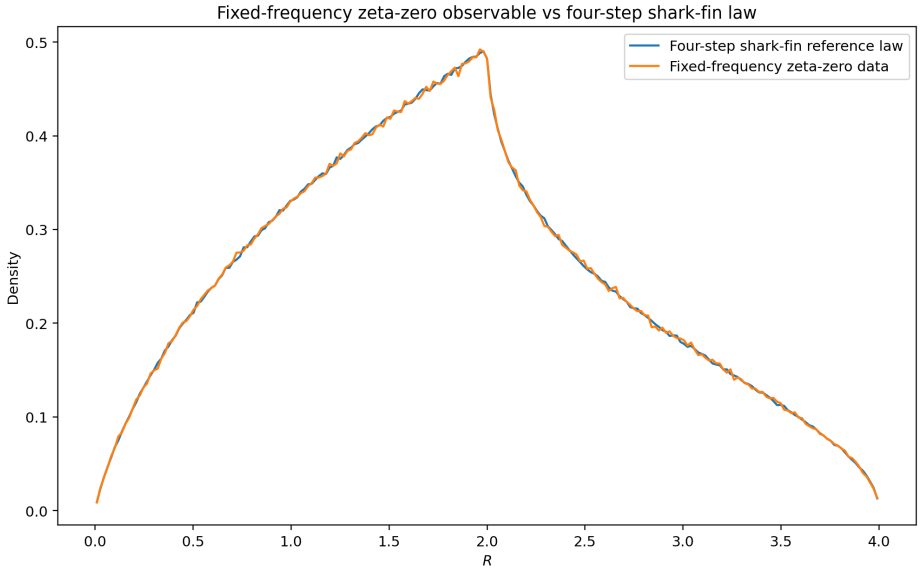


Fig. 2 Fixed-frequency four-phase zeta-zero observable for $u = (1, \sqrt{2}, \sqrt{3}, \sqrt{5})$, compared with an independent simulation of the classical four-step radial reference law.

7.2 Lagged $q_k = k \log k$ diagnostic

Define

$$q_k = k \log k, \qquad z_k = e^{i\gamma_k q_k}, \qquad w_k = e^{i\tau_k q_k},$$

and

$$\Delta_{1,1}(k) = (z_{k+1} - z_k) - (w_{k+1} - w_k).$$

This uses adjacent indices and a frequency depending on k. Starting at $k = 3$ leaves 2,001,049 evaluable terms from the full 2,001,052-entry arrays; no shorter input dataset is used. It is therefore not an instance of Theorem 7.4. Its raw diagnostics are:

Quantity	Value
sample count	$2,001,049$
mean radius, data	1.80020415689
exact shark-fin mean	1.799092479843
mean deviation	$1.111677e-03$
second moment, data	4.004212467
exact second moment	4
second-moment deviation	$4.212467e-03$
normalized-shape KS diagnostic	$5.41e-04$

The shape is close to the four-step radial profile, but the raw first and second moments show a finite systematic offset. Normalizing each sample by its own empirical mean removes a scale parameter and therefore makes the corresponding KS statistic a shape diagnostic, not an exact distribution-free two-sample test.

The source ordinate table is quoted with accuracy within 4×10^{-9}. Since $q_k = k \log k$, this accuracy is amplified in the phase. At the top of the dataset the worst-case phase uncertainty from that quoted bound is of order 0.1 radian. This lagged high-frequency diagnostic is therefore tied to the supplied ordinate table and is not used as an asymptotic theorem check. Figure 3 shows its normalized radial profile.

8 Adjacent-zero diagonal limit and growing-window rigidity

The ordering of the zeros produces a rigidity statement below the scale treated by Kargin's separated-index covariance theorem. The proof uses only the one-point prefix law, slow variation of the smooth clock and normalization, and the monotonicity $\gamma_{k+j} \geq \gamma_k$.

Theorem 8.1 (Growing-window collapse and decoupling) *Let r_N be a nonnegative integer sequence satisfying*

$$r_N = o\Big(\sqrt{\log\log N}\Big). \tag{8.1}$$

Let K_N be uniform on

$$\{k_0, \dots, N - r_N\}.$$

Then the following statements hold unconditionally.

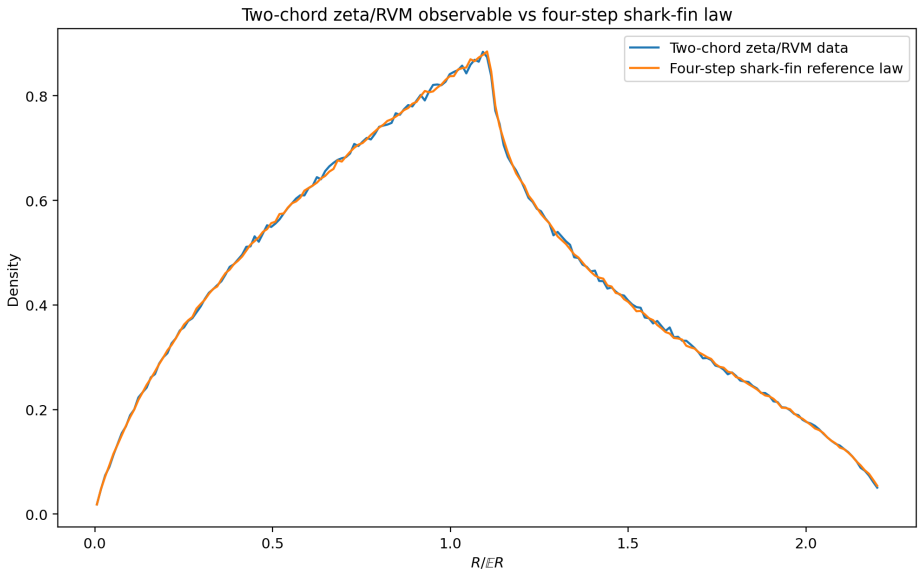


Fig. 3 Lagged two-chord zeta/Riemann–von Mangoldt observable compared with the four-step radial reference law after the normalization described in the text. This is a diagnostic for the supplied finite table, not an instance of Theorem 7.4.

(i)
$$\max_{0\le j\le r_N} |X_{K_N+j} - X_{K_N}| \longrightarrow 0 \tag{8.2}$$
in probability.

(ii) *For every fixed integer* $s \ge 0$,
$$(X_{K_N}, X_{K_N+1}, \dots, X_{K_N+s}) \Longrightarrow (G, G, \dots, G), \tag{8.3}$$
where all coordinates on the right are the same standard Gaussian variable. Moreover, for every fixed $j \ge 1$,
$$\operatorname{Corr}(X_{K_N}, X_{K_N+j}) \longrightarrow 1. \tag{8.4}$$

(iii) *For every fixed integer* $s \ge 0$, *every* $J \ge 1$, *and every* $u \in \mathcal{U}_J$ *from Theorem 5.3,*
$$(X_{K_N}, \dots, X_{K_N+s}, (u_1\gamma_{K_N}, \dots, u_J\gamma_{K_N}) \bmod 2\pi) \Longrightarrow (G, \dots, G, \Theta), \tag{8.5}$$
where Θ *is Haar-uniform on* $\mathbb{T}^J$ *and independent of* G.

Proof Fix $\epsilon \in (0,1)$. On the bulk event $K_N \ge \epsilon N$, whose probability tends to at least $1-\epsilon$, the clock relation and the mean-value theorem give, uniformly for $0 \le i < j \le r_N$,
$$\tau_{K_N+j} - \tau_{K_N+i} \ll \frac{j-i}{\log N}. \tag{8.6}$$
The normalization is slowly varying on the same range:
$$\max_{0\le i,j\le r_N} \left| \frac{\sigma_{K_N+i}}{\sigma_{K_N+j}} - 1 \right| = o(1). \tag{8.7}$$

Also

$$\min_{0\le j\le r_N} \sigma_{K_N+j} \asymp \frac{\sqrt{\log\log N}}{\log N}. \tag{8.8}$$

Hence, by (8.1),

$$b_N(\epsilon) := \max_{\substack{0\le i<j\le r_N \\ K_N\ge \epsilon N}} \frac{\tau_{K_N+j}-\tau_{K_N+i}}{\min_{0\le m\le r_N}\sigma_{K_N+m}} \longrightarrow 0. \tag{8.9}$$

For $0 \le i < j \le r_N$, zero ordering gives

$$\sigma_{K_N+j}X_{K_N+j} - \sigma_{K_N+i}X_{K_N+i} \ge -(\tau_{K_N+j}-\tau_{K_N+i}). \tag{8.10}$$

Taking $i = 0$ and $j = r_N$,

$$X_{K_N+r_N} \ge \frac{\sigma_{K_N}}{\sigma_{K_N+r_N}} X_{K_N} - b_N(\epsilon). \tag{8.11}$$

The variables X_{K_N} are tight by Lemma 3.4. The shifted variables $X_{K_N+r_N}$ have the same limiting marginal. Indeed, for every fixed moment order q, shifting a prefix by $r_N = o(N)$ changes its average by at most

$$\frac{2r_N}{N}\max_{k\le N+r_N}|X_k|^q \ll \frac{r_N\log^q(2N)}{N} = o(1),$$

because here $r_N = o(\sqrt{\log\log N})$ and Lemma 3.3 applies. Thus $X_{K_N+r_N}$ is tight and also converges to $N(0,1)$. By (8.7)–(8.11), the negative part of $X_{K_N+r_N} - X_{K_N}$ tends to zero in probability.

Consider a weakly convergent subsequence of the endpoint pair and write its limit as (Y_1, Y_2). Portmanteau applied to $\{(x,y) : y - x < -\delta\}$ gives $Y_2 \ge Y_1$ almost surely. Both coordinates have the same $N(0,1)$ law. More generally, if two real random variables have the same law and $Y_2 \ge Y_1$ almost surely, then they are equal almost surely: for every rational q,

$$\{Y_1 > q\} \subseteq \{Y_2 > q\}$$

with equal probabilities, and

$$\{Y_1 < Y_2\} = \bigcup_{q\in\mathbb{Q}} \{Y_1 \le q < Y_2\}$$

is null. Thus every subsequential endpoint limit is (G, G), and

$$X_{K_N+r_N} - X_{K_N} \longrightarrow 0 \tag{8.12}$$

in probability.

For an intermediate $j \le r_N$, (8.10) with $(i,j) = (0,j)$ gives a lower bound by $X_{K_N} - o_{\mathbb{P}}(1)$, uniformly in j. Applying the same inequality to $(i,j) = (j, r_N)$ gives an upper bound by $X_{K_N+r_N} + o_{\mathbb{P}}(1)$, uniformly in j. Together with (8.12), this proves (8.2). Since ϵ was arbitrary, the bulk restriction disappears.

For fixed s, (8.3) follows from (8.2), Lemma 3.4, and Slutsky. The uniform fourth moment bound in Lemma 3.4, applied also to every fixed shift, gives uniform integrability of the squares and cross-products; therefore the second mixed moments pass to the limit and yield (8.4).

Finally Theorem 5.3 is unchanged if the uniform prefix is truncated by $r_N = o(N)$ indices. For fixed s, (8.2) gives

$$(X_{K_N}, \dots, X_{K_N+s}, \Theta_{K_N}) = (X_{K_N}, \dots, X_{K_N}, \Theta_{K_N}) + o_{\mathbb{P}}(1),$$

where

$$\Theta_{K_N} := (u_1\gamma_{K_N}, \dots, u_J\gamma_{K_N}) \bmod 2\pi.$$

Slutsky and Theorem 5.3 prove (8.5). □

8.1 Relation to Kargin's separated-index theorem

Kargin's bivariate theorem [7, Thm. 1.2] assumes

$$k' - k \asymp (\log k)^{\beta}, \qquad \beta > 0,$$

and gives a bivariate Gaussian limit with correlation $(1-\beta)_+$. Consequently, it does not contain the bounded-gap case $\beta = 0$. Since

$$\sqrt{\log\log N} = o\big((\log N)^{\beta}\big) \qquad (\beta > 0 \text{ fixed}),$$

the range $r_N = o(\sqrt{\log\log N})$ of Theorem 8.1 lies strictly below every separation scale covered by Kargin's joint theorem.

What is new here beyond Kargin. The additional conclusions established here are:

1. for every fixed $r \geq 1$,
$$(X_{K_N}, X_{K_N+r}) \Longrightarrow (G, G),$$
so the full first-order joint law is obtained at $\beta = 0$;
2. the fixed-gap statement is strengthened to simultaneous collapse over every growing window $r_N = o(\sqrt{\log\log N})$;
3. the resulting bounded-gap block inherits the same-zero phase decoupling (8.5); and
4. the proof uses zero ordering, the smooth-clock scale, tightness, and the same-law order argument rather than an extension of Kargin's positive-power prime-sum moment calculation.

Thus the new content relative to [7] is the joint diagonal law at the excluded $\beta = 0$ edge, its growing-window extension, and its phase-decoupled form.

Why the prime-sum input does not fill the gap. Kargin's covariance calculation invokes Bourgade's mesoscopic estimate [8]. Bourgade's result is a statement about a prime sum, and its bounded-shift regime is compatible with the formal value 1 at the $\beta = 0$ scale. It is not a theorem about $\mathrm{Corr}(X_k, X_{k+r})$, nor does it provide a joint law for bounded index separation. Transport from that analytic input to zero fluctuations in Kargin's paper uses the joint-moment and comparison machinery formulated under the hypothesis $\beta > 0$. The present theorem supplies a different route to the bounded-gap joint law and makes no claim that the prime-sum estimate alone already contained that result.

Remark 8.2 (Critical-scale interpretation) Theorem 8.1 proves collapse below $\sqrt{\log\log N}$. If the canonical increment

$$D_k := \sqrt{\log\log \tau_k}\,(X_{k+1} - X_k)$$

has a nondegenerate local limit, then this is also the natural scale at which accumulated microscopic increments can become order one. This observation motivates the scale but is not used as a proof of optimality of (8.1).

9 Exact bridge from displacement increments to nearest-neighbor gaps

Put

$$L_k := \log \frac{\tau_k}{2\pi}, \qquad d_k^{\mathrm{clk}} := \frac{2\pi}{L_k}.$$

The clock equation and the mean-value theorem give

$$\tau_{k+1} - \tau_k = d_k^{\mathrm{clk}}(1 + o(1)).$$

Write $\sigma(T) = \sqrt{2 \log\log T}/\log(T/(2\pi))$. Direct differentiation gives

$$\frac{\sigma'(T)}{\sigma(T)} = O\left(\frac{1}{T \log(T/(2\pi))}\right).$$

Since $\tau_{k+1} - \tau_k = O(1/L_k)$, another application of the mean-value theorem yields

$$\frac{|\sigma_{k+1} - \sigma_k|}{d_k^{\mathrm{clk}}} = O\left(\frac{\sigma_k}{\tau_k L_k}\right) = o(1).$$

Moreover, Lemma 3.3 gives $|X_{k+1}| \ll \log(2+k)$. Hence

$$\frac{(\sigma_{k+1} - \sigma_k)X_{k+1}}{d_k^{\mathrm{clk}}} = o(1) \tag{9.1}$$

pointwise as $k \to \infty$. In particular this error is negligible in probability and in every fixed local second-moment calculation used below. Using the exact decomposition

$$\sigma_{k+1}X_{k+1} - \sigma_k X_k = \sigma_k(X_{k+1} - X_k) + (\sigma_{k+1} - \sigma_k)X_{k+1},$$

we obtain

$$\frac{\gamma_{k+1} - \gamma_k}{d_k^{\mathrm{clk}}} = \frac{\tau_{k+1} - \tau_k}{d_k^{\mathrm{clk}}} + \frac{\sigma_k}{d_k^{\mathrm{clk}}}(X_{k+1} - X_k) + o_{\mathbb{P}}(1) \tag{9.2}$$

$$= 1 + \frac{\sqrt{2 \log\log \tau_k}}{2\pi}(X_{k+1} - X_k) + o_{\mathbb{P}}(1). \tag{9.3}$$

Thus the normalized gap is governed by

$$D_k := \sqrt{\log\log \tau_k}\,(X_{k+1} - X_k), \tag{9.4}$$

not by the first-order limit of either X_k separately. Equation (9.3) can be written

$$G_k^{\mathrm{gap}} = 1 + \frac{\sqrt{2}}{2\pi} D_k + o_{\mathbb{P}}(1). \tag{9.5}$$

Thus the microscopic displacement increment and the normalized nearest-neighbor gap are related, to first order, by the explicit affine map $D \mapsto 1 + D/(\sqrt{2}\pi)$. Nonnegativity of the normalized gap corresponds at leading order to the boundary

$$D = -\sqrt{2}\,\pi = -4.44288294\ldots.$$

This is an asymptotic dictionary rather than a pointwise support statement, because the remainder in (9.5) is $o_{\mathbb{P}}(1)$.

Corollary 9.1 (Transfer of any eventual microscopic gap law) *Assume that the prefix law of D_k converges to a moment-determinate probability law ν_D, that every fixed moment converges, that the corresponding fixed absolute moments are uniformly bounded, and that the algebraic polynomials are dense in $L^2(\nu_D)$. Then for every $J \geq 1$ and Lebesgue-almost every $u \in \mathbb{R}^J$,*

$$(D_k, (u_1\gamma_k, \ldots, u_J\gamma_k) \bmod 2\pi) \Longrightarrow (D, \Theta), \tag{9.6}$$

where $D \sim \nu_D$, Θ is Haar-uniform on $\mathbb{T}^J$, and the two factors are independent.

Proof Apply Proposition 5.5 with $\Psi_k = D_k$. □

Remark 9.2 Corollary 9.1 does not supply the law of D_k; its hypotheses contain precisely the missing microscopic spacing information. Its content is a transfer statement: once such a law is established, no new twisted-moment calculation is required to decouple it from almost every fixed arithmetic phase vector.

Proposition 9.3 (Variance-correlation relation at the microscopic scale) *Let*

$$\mathcal{I}_n = \{a_n, a_n + 1, \ldots, b_n\}, \qquad a_n \longrightarrow \infty,$$

be consecutive index windows, and let K_n be uniform on $\mathcal{I}_n^- := \{a_n, \ldots, b_n - 1\}$. Choose any representative $k_n \in \mathcal{I}_n$ and put

$$\ell_n := \log\log \tau_{k_n}.$$

Assume

$$\sup_{k \in \mathcal{I}_n} \left| \frac{L_k}{L_{k_n}} - 1 \right| \longrightarrow 0, \qquad \delta_n := \sup_{k \in \mathcal{I}_n} |\log\log \tau_k - \ell_n| \longrightarrow 0. \tag{9.7}$$

All variances and correlations below are taken with respect to the uniform law of K_n. Write

$$G_{K_n}^{\mathrm{gap}} := \frac{\gamma_{K_n+1} - \gamma_{K_n}}{2\pi/L_{K_n}}, \qquad c_n := \frac{\sqrt{2\ell_n}}{2\pi},$$

and set

$$V_{0,n} := \mathrm{Var}(X_{K_n}), \qquad V_{1,n} := \mathrm{Var}(X_{K_n+1}), \qquad \rho_{1,n} := \mathrm{Corr}(X_{K_n}, X_{K_n+1}).$$

Suppose

$$V_{0,n} \longrightarrow v, \qquad V_{1,n} \longrightarrow v, \qquad \mathbb{E}X_{K_n}^2 + \mathbb{E}X_{K_n+1}^2 = O(1)$$

for some $v > 0$, and assume the shifted-window variance imbalance is negligible on the microscopic scale:

$$\ell_n \big(\sqrt{V_{0,n}} - \sqrt{V_{1,n}}\big)^2 \longrightarrow 0. \tag{9.8}$$

Then

$$\mathrm{Var}\big(G^{\mathrm{gap}}_{K_n}\big) = \frac{v\,\ell_n}{\pi^2}\,(1-\rho_{1,n})\,\big(1+o(1)\big) + o(1). \tag{9.9}$$

Consequently, if the normalized gap variance tends to a finite nonzero limit V, then

$$1-\rho_{1,n} = \frac{\pi^2 V}{v\,\ell_n} + o\!\left(\frac{1}{\ell_n}\right).$$

The asymptotic unit-variance case is $v \to 1$; equation (9.9) is also the finite-height algebraic relation used in the numerical reconstructions below.

Remark 9.4 (Second moment in place of variance) In the numerical reconstructions the measured second moment $\langle X^2\rangle$ is used in place of the variance. Over the full prefix, $\langle X\rangle = -3.48\times10^{-7}$ and hence $\langle X\rangle^2 = 1.2\times10^{-13}$. On the top-window set of 250,001 zero indices used below, the corresponding square of the mean is 3.06×10^{-11}. Both are far below every digit reported in the variance comparisons.

Proof By (9.3) and (9.1), for $k \in \mathcal{I}_n^-$,

$$G^{\mathrm{gap}}_k = 1 + \frac{\sqrt{2\log\log\tau_k}}{2\pi}\,(X_{k+1}-X_k) + \eta_{k,n}. \tag{9.10}$$

Here the rate can be read directly from the same clock estimates, without strengthening (9.3). Since $M'(T) = L(T)/(2\pi)$ and $M''(T) = 1/(2\pi T)$, the mean-value theorem gives

$$\frac{\tau_{k+1}-\tau_k}{2\pi/L_k} - 1 = O\!\left(\frac{1}{\tau_k L_k^2}\right).$$

Moreover, the estimate preceding (9.1) and Lemma 3.3 give

$$\frac{(\sigma_{k+1}-\sigma_k)X_{k+1}}{2\pi/L_k} = O\!\left(\frac{\sigma_k\log(2+k)}{\tau_k L_k}\right) = O\!\left(\frac{\sqrt{\log\log\tau_k}}{k}\right),$$

where the last step uses $\tau_k \asymp k/\log k$ and $L_k \asymp \log k$. Consequently, uniformly for $k \ge a_n$,

$$\eta_{k,n} = O\!\left(\frac{\sqrt{\log\log\tau_k}}{k}\right).$$

By (9.7), $\log\log\tau_k = \ell_n + o(1)$ uniformly on $\mathcal{I}_n$, so

$$\sqrt{\ell_n}\sup_{k\in\mathcal{I}_n^-}|\eta_{k,n}| = O\!\left(\frac{\ell_n}{a_n}\right) = o(1). \tag{9.11}$$

Thus the additive remainder is small at the rate needed for a local variance calculation; no supremum bound on $X_{k+1}-X_k$ is used.

Factor the slowly varying coefficient as

$$\frac{\sqrt{2\log\log\tau_k}}{2\pi} = c_n\big(1+\theta_{k,n}\big), \qquad \theta_{k,n} := \sqrt{\frac{\log\log\tau_k}{\ell_n}} - 1. \tag{9.12}$$

Since $\delta_n \to 0$ and $\ell_n \to \infty$,

$$\Theta_n := \sup_{k\in\mathcal{I}_n}|\theta_{k,n}| = O\!\left(\frac{\delta_n}{\ell_n}\right). \tag{9.13}$$

Write

$$A_n := c_n(X_{K_n+1} - X_{K_n}), \qquad R_n := \theta_{K_n,n}A_n + \eta_{K_n,n}.$$

The bounded local second moments imply $\mathbb{E}A_n^2 = O(\ell_n)$. Hence

$$\|\theta_{K_n,n} A_n\|_2 \le \Theta_n \|A_n\|_2 = O\left(\frac{\delta_n}{\sqrt{\ell_n}}\right),$$

while (9.11) gives $\sqrt{\ell_n}\,\|\eta_{K_n,n}\|_2 \to 0$. Therefore

$$\sqrt{\ell_n}\,\|R_n\|_2 \longrightarrow 0. \tag{9.14}$$

Since $\mathrm{Var}(A_n)^{1/2} \le \|A_n\|_2 = O(\sqrt{\ell_n})$, Cauchy–Schwarz yields

$$\left|\mathrm{Var}(1 + A_n + R_n) - \mathrm{Var}(A_n)\right| \le 2\,\mathrm{Var}(A_n)^{1/2}\,\mathrm{Var}(R_n)^{1/2} + \mathrm{Var}(R_n) = o(1). \tag{9.15}$$

This is the required second-moment control: the coefficient error is handled in L^2 and the additive clock error has an explicit rate.

It remains to evaluate the main variance. Exactly,

$$\mathrm{Var}(A_n) = c_n^2\left(V_{0,n} + V_{1,n} - 2\rho_{1,n}\sqrt{V_{0,n}V_{1,n}}\right) \tag{9.16}$$

$$= \frac{\ell_n}{\pi^2}\sqrt{V_{0,n}V_{1,n}}\,(1 - \rho_{1,n}) + \frac{\ell_n}{2\pi^2}\left(\sqrt{V_{0,n}} - \sqrt{V_{1,n}}\right)^2. \tag{9.17}$$

By (9.8) the second term is $o(1)$, while $\sqrt{V_{0,n}V_{1,n}} = v(1+o(1))$. Combining (9.15) and (9.17) proves (9.9). □

Remark 9.5 (Numerical verification of the window hypotheses) For the top 250,000 consecutive pairs used in Section 9.1, take $\mathcal{I}$ to be the 250,001 supporting zero indices and k_n its midpoint. Directly from the canonical data,

$$\sup_{k\in\mathcal{I}}\left|\frac{L_k}{L_{k_n}} - 1\right| = 0.005249, \qquad \sup_{k\in\mathcal{I}}\left|\log\log\tau_k - \log\log\tau_{k_n}\right| = 0.004564.$$

The variance-balance term in (9.8) is 2.45×10^{-10}. With the midpoint coefficient, (9.17) gives 0.167466, against the measured normalized-gap variance 0.167463. If instead the actual window-average coefficient $\overline{\log\log\tau} = 2.630452547$ is retained, the same local identity gives 0.167462766, compared with the measured 0.167462770. The difference between the midpoint and averaged evaluations is exactly the finite-window variation quantified above.

Remark 9.6 (What is and is not inherited from separated-index analysis) At the prime-sum input level, Bourgade's bounded-shift estimate [8] is compatible with the formal $\beta = 0$ value that appears inside Kargin's covariance calculation. It is not a theorem about $\mathrm{Corr}(X_k, X_{k+r})$ at bounded index separation and it does not supply a joint zero law. Theorem 8.1 obtains that first-order joint law by a different mechanism. The nearest-neighbor spacing information then survives only in the correction to the diagonal, at order $1/\sqrt{\log\log\tau_k}$ for displacement increments and $1/\log\log\tau_k$ for covariance.

9.1 Finite-height adjacent-zero audit

Using all 2,001,051 consecutive pairs from the same ordinate table gives

Quantity	Value
$\langle X^2 \rangle$, full prefix	0.848888
$\mathrm{Corr}(X_k, X_{k+1})$, full prefix	0.246643
$\langle D_k \rangle$, full prefix	-1.64×10^{-7}
$\mathrm{Var}(D_k)$, full prefix	3.278190
$\min D_k$, full prefix	-4.418684
mean normalized gap	0.999999
variance normalized gap	0.166075

The observed full-prefix minimum,

$$\min_k D_k = -4.418684,$$

lies close to the leading-order boundary $-\sqrt{2}\pi = -4.442883\ldots$ furnished by (9.5). This is only a finite-height diagnostic: the $o_{\mathbb{P}}(1)$ remainder in the gap dictionary does not promote that leading-order boundary to a pointwise support theorem.

For the top 250,000 consecutive pairs extracted from the same full 2,001,052-entry arrays, where the slowly varying coefficients are essentially constant on the comparison scale, the reported marginal second moment is the unweighted mean over the 250,001 distinct zero indices supporting those pairs:

Quantity	Value
median τ	1.067435×10^6
mean $\log\log\tau$	2.630453
mean marginal $\langle X^2 \rangle$	0.856734
empirical ρ_1	0.266593
empirical normalized-gap variance	0.167463
ρ_1 using Wigner–GUE variance	0.220025

9.2 Finite-height GUE benchmark

For the Wigner surmise in symmetry class GUE,

$$V_{\mathrm{GUE}}^{\mathrm{W}} = \frac{3\pi}{8} - 1 = 0.178097\ldots, \tag{9.18}$$

whereas the GOE Wigner variance is

$$V_{\mathrm{GOE}}^{\mathrm{W}} = \frac{4}{\pi} - 1 = 0.273240\ldots.$$

Thus a variance near 0.286 must not be used as a GUE benchmark. The finite-height comparison below uses throughout the GUE Wigner-surmise value $3\pi/8 - 1 = 0.178097\ldots$; the corresponding GOE value is $4/\pi - 1 = 0.273240\ldots$.

Combining Proposition 9.3 with (9.18), and using asymptotic unit variance for X, gives the benchmark

$$\rho_1(T) \approx 1 - \frac{\pi^2 V_{\mathrm{GUE}}^{\mathrm{W}}}{\log\log T}.$$

Numerically,

$$\rho_1(10^6) = 0.330583\ldots, \qquad \rho_1(10^{12}) = 0.470388\ldots, \qquad \rho_1(10^{30}) = 0.584970\ldots.$$

If the measured finite-height second moment $\langle X^2\rangle = 0.848888$ is inserted into the finite-height algebraic relation, the corresponding values are $0.211419\ldots$, $0.376111\ldots$, and $0.511089\ldots$.

The empirical top-window normalized-gap variance 0.167463 is about 6.0% below the Wigner–GUE benchmark $0.178097\ldots$. At the present height this is recorded only as a finite-height discrepancy; no convergence rate to a GUE spacing law is asserted. Figure 4 shows the corresponding finite-height correlation benchmark.

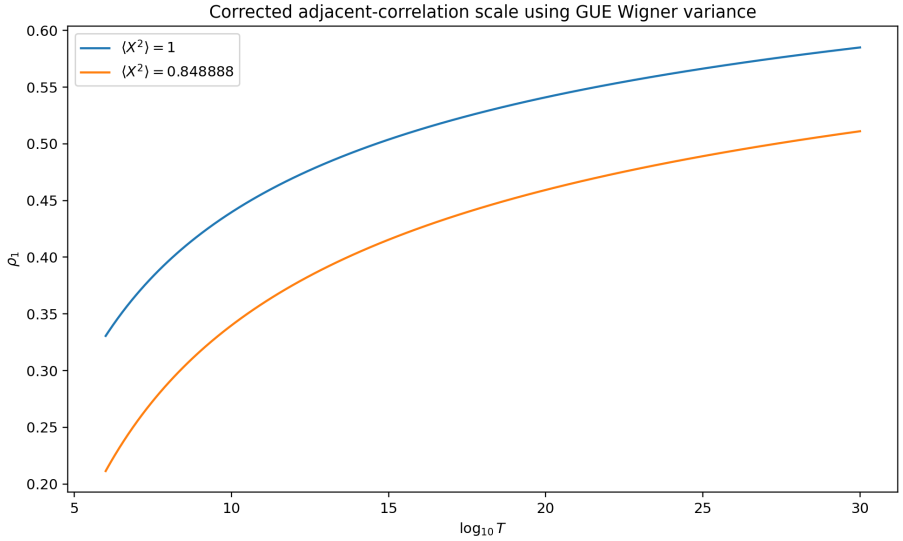


Fig. 4 Finite-height adjacent-correlation benchmark based on the corrected Wigner–GUE spacing variance. The two curves distinguish asymptotic unit variance for X from the measured finite-height value 0.848888.

10 Reproducibility and data provenance

The zero ordinates used throughout this article are the first 2,001,052 nontrivial Riemann-zeta ordinates published by A. M. Odlyzko in his public tables of zeta zeros [22]. Odlyzko's table states an accuracy within 4×10^{-9} for this dataset. The original plain-text table is included verbatim in the repository as `data/zeros6.txt`, so the primary numerical source accompanies the manuscript. The file `data/zeross.bin` is

the deterministic double-precision binary encoding of those same ordinates used by the numerical scripts. The file `data/zLamb.bin` is deterministic derived data: it is the smooth Riemann–von Mangoldt clock table computed from those ordinates and the clock definition used in this paper. These are the only three data files required by the article. Every numerical script first loads and verifies the full 2,001,052-entry `zeross.bin` and `zLamb.bin` arrays. Any smaller sample count reported in a derived calculation is forced by the mathematical observable itself—for example $N-1$ adjacent pairs, $N-j$ pairs at lag j, or an explicitly labelled top-height window—and never comes from a separate or truncated input dataset.

Reproducibility. All numerical tables and figures in this version are generated from the supplied `zeross.bin` and `zLamb.bin` datasets by the scripts distributed with the source archive. The archive includes the generated CSV/TXT outputs and a SHA256 manifest. No numerical value used in a theorem proof is taken from these computations.

11 Use of AI tools

Codex was used to check English usage and LaTeX formatting, to independently double-check computations in Python, and to generate reproducibility scripts and figures. Claude was used to identify potential inconsistencies and to suggest independent numerical checks. All mathematical claims and numerical values included in the final manuscript were independently audited against the stated definitions and cited literature by the author, who assumes sole responsibility for the content.

12 Literature position and scope

For the fixed-frequency shark-fin theorem, Hlawka [3] supplies the one-dimensional Weyl tests and Weyl's criterion [13] assembles the torus vector. The resulting deterministic zeta-zero radial law coincides with the classical four-step radial density studied in the short-walk literature [10, 11].

The same-zero Gaussian–phase coupling is different from simply juxtaposing Kargin's Gaussian marginal [7] and Hlawka's phase marginal [3]. Closely related conditional same-index architectures do occur in Çiçek–Gonek [15]: under RH and a zero-spacing hypothesis they combine a Gaussian statistic attached to a zero with uniform distribution of the selected ordinate subsequence. Their twisted analysis includes sums in which an exponential phase is weighted by powers of a Dirichlet polynomial. The present result has a different tradeoff: it is unconditional and applies to Lebesgue-almost every fixed frequency vector, whereas Çiçek–Gonek obtain conditional results at prescribed frequencies. Their work builds on Çiçek's distribution theorem for the logarithm of ζ near nontrivial zeros [14]. The metric sinc-kernel argument here is therefore positioned as an unconditional almost-every-frequency decoupling principle, not as the first appearance of a Gaussian-statistic/equidistribution architecture.

The adjacent-zero theorem introduces a genuine two-zero object but shows that its first-order normalized displacement limit is completely degenerate:

$$(X_k, X_{k+1}) \Rightarrow (G, G).$$

The local spacing information begins only after magnifying the difference by $\sqrt{\log\log\tau_k}$. Equivalently, the first nontrivial covariance information is the coefficient of

$$\frac{1}{\log\log\tau_k}$$

in $1 - \mathrm{Corr}(X_k, X_{k+1})$.

Unnormalized exponential sums at prime-power logarithms exhibit the classical Landau/von-Mangoldt resonance. They do not produce exceptional frequencies for Hlawka's normalized equidistribution. The genuinely unresolved twisted objects in the present metric argument are the displacement-weighted sums at prescribed arithmetic frequencies,

$$\frac{1}{N}\sum_{k\le N} X_k^{\ell} e^{iv\gamma_k}, \qquad \ell \ge 1,$$

when v is fixed in advance rather than chosen from the full-measure set supplied by Theorem 4.2.

13 Conclusion

The main result is the same-zero theorem. We also record an adjacent-zero diagonal consequence of the ordinal viewpoint.

The same-zero theorem states that, for almost every fixed frequency vector, Kargin's normalized displacement becomes asymptotically independent of the torus phases of the same zero. The metric sinc-kernel argument supplies the missing mixed cancellation and yields the Gaussian–phase product law.

The adjacent-zero diagonal consequence states that consecutive normalized displacements have the joint first-order limit (G, G). This identifies exactly why one-zero marginal information cannot determine nearest-neighbor spacing. The spacing law resides in

$$\sqrt{\log\log\tau_k}\,(X_{k+1} - X_k),$$

or equivalently in the $1/\log\log\tau_k$ correction to adjacent correlation.

Although the shark-fin law is an immediate consequence of Haar measure on $\mathbb{T}^4$, the main objective of the present work is the same-zero Gaussian–phase coupling and the metric independence theorem from which such radial laws arise.

The four-frequency shark-fin result is retained as a deterministic zeta-zero realization of the classical four-step radial law. Its proof uses Hlawka, Weyl, and the continuous two-chord map, and does not assume any random-walk model for the zeros. In the classical short-walk literature the same radial density is obtained from a random-walk construction; the zeta-zero result establishes it without that hypothesis. It is noteworthy that this classical radial law appears in a deterministic zeta-zero observable, where no random-walk structure of the zeros is assumed or known.

Declarations

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Conflict of interest

The author declares that there is no conflict of interest.

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